Securing AI and Advanced Topics

Today's goal • 3/3 items

Module 1
Course Introduction

Course Overview
Reading • 10 min

Instructor Biography - Lanier Watkins
Reading • 2 min

Module 2
Fraud Prevention with Cloud AI Solutions

Credit Card Fraud Threats and AI Prevention

Credit Card Fraud Prevention with AI
Video • 4 min

Reading References
Reading • 10 min

Credit Card Fraud Threats and AI Prevention
Practice Assignment • Grade: 100%

Implementing and Evaluating IBM Watson for Fraud Detection

Credit Card Fraud Prevention: IBM Watson Example
Video • 10 min

Reading References
Reading • 10 min

Implementing and Evaluating IBM Watson for Fraud Detection
Practice Assignment • Grade: 100%

Module-end Assessments

Self-Reflective Reading: Understanding of AI Fraud Prevention Tools
Reading • 45 min

Fraud Prevention with Cloud AI Solutions
Graded Assignment • 1h

Module 3
Introduction to Generative Adversarial Attacks (GANs)

Module 4
GANs and Adversarial Attacks

Module 5
Reinforcement Learning

Module 6
Evaluating AI Models and Performance

Self-Reflective Reading: Understanding of AI Fraud Prevention Tools

0:10 / 3:20 2x

Introduction

In the digital age, the threat of fraud has become increasingly sophisticated, necessitating the development of advanced fraud prevention tools powered by Artificial Intelligence (AI). The module "Fraud Prevention with Cloud AI Solutions" explores various AI tools that organizations can leverage to mitigate fraudulent activities effectively. This literature review aims to examine the current landscape of AI fraud prevention tools, comparing class examples with state-of-the-art solutions. Additionally, we will analyze how accurate AI fraud prevention tools can streamline the work of cyber analysts. By using the language from the machine learning (ML) analytic development process, we will summarize the development of each ML analytic example mentioned in the IBM Watson lecture. Finally, we will investigate the performance tuning parameters of classifiers discussed in the lecture, optimize them, and evaluate their performance.

Tasks

1. **Literature Review on AI Fraud Prevention Tools**:
   - Research various AI tools used for fraud prevention.
   - Compare the class example with the latest tools available in the market.
2. **Impact on Cyber Analysts**:
   - Describe how an accurate AI fraud prevention tool can facilitate the work of cyber analysts.
3. **ML Analytic Development Process**:
   - Summarize the development of each ML example analytic mentioned in the IBM Watson lecture using appropriate ML language.
4. **Performance Tuning**:
   - Investigate the Python documentation for the classifiers discussed in the IBM Watson lecture.
   - Identify performance tuning parameters for each classifier.
   - Optimize these parameters to improve performance and report the results.

Advised Steps

1. **Conduct a Literature Review**:
   - Start by exploring academic journals, articles, and case studies related to AI fraud prevention tools.
   - Document the key features, advantages, and limitations of various tools.
2. **Analyze Impact on Cyber Analysts**:
   - Reflect on the daily tasks of cyber analysts and how AI can enhance their efficiency and accuracy.
3. **Summarize ML Analytic Development**:
   - Review the lecture materials and summarize the steps involved in developing each ML analytic using the specified terminology.
4. **Explore Python Documentation**:
   - Use Jupyter Notebook to access the classifiers' documentation.
   - Analyze and identify key performance tuning parameters, then optimize and test them.

Self-Reflective Questions

1. **Literature Review**:
   - What key features differentiate the AI fraud prevention tools I researched?
   - How do these tools compare with the class example in terms of functionality and effectiveness?
2. **Impact on Cyber Analysts**:
   - In what specific ways would an accurate AI fraud prevention tool improve my ability to perform cyber analysis?
   - How can I leverage AI tools to enhance my decision-making process in fraud detection?
3. **ML Analytic Development**:
   - How do the steps in the ML analytic development process relate to real-world applications in fraud prevention?
   - What challenges might arise during the implementation of these ML analytics?
4. **Performance Tuning**:
   - What performance tuning parameters had the most significant impact on the classifiers' accuracy?
   - How did my optimizations influence the overall effectiveness of the AI fraud prevention tools?

Go to next item ✓ Completed

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